



Fig. 3: Warning message on LCD

PARTS LIST

Semiconductors:

Board1 - Arduino Uno board
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 220-ohm
VR1 - 10-kilo-ohm

Miscellaneous:

BATT.1 - 9V battery
PZ1 - Piezo buzzer
CON1 - 3-pin Berg connector for IR proximity module
LCD1 - 16x2 alphanumeric LCD

tal oscillator, USB connection, power jack, ICSP header and reset button. Output of the sensor module is ana-

logue. It gets converted to digital through the inbuilt A/D converter of Arduino Uno.

Preset. A 10-kilo-ohm preset (VR1) is a three-legged electronic component used to vary the resistance in a circuit. Here, it is used to control the contrast of the LCD.

LCD. A 16-pin JHD162A LCD module is used in this project as a warning display. It is a two-lines and 16-characters LCD module, which can be replaced with a large LCD display (with some modifications in the circuit) for actual use.

Programming of Arduino Uno is done using Arduino IDE software. The program (vehicle_detection.ino) is written in Arduino programming language.

Construction and testing

Make connections as per the circuit diagram. If an object or vehicle is not detected by the sensor, 'Path is clear'

EFY Note
The source code of this project is available for free download at source.efymag.com

message is displayed on LCD1. If a vehicle is detected, the message reads as 'Please Stop!!' At the same time LED1 glows and the buzzer sounds an alarm (Fig. 3). The circuit can be wired on a breadboard with jumper wires for testing.

This system can be implemented using a laser sensor, speaker and bigger display, instead of a proximity sensor, buzzer and 16x2 LCD, respectively. The sensor and display should be installed along the path of the vehicle, before it approaches the bridge, so that the driver gets notified of the warning message well in time. **EFY**



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